



American International University-Bangladesh (AIUB)

Department of Computer Science

Course Name: **COMPUTER GRAPHICS**

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Course Code: CSE 4118

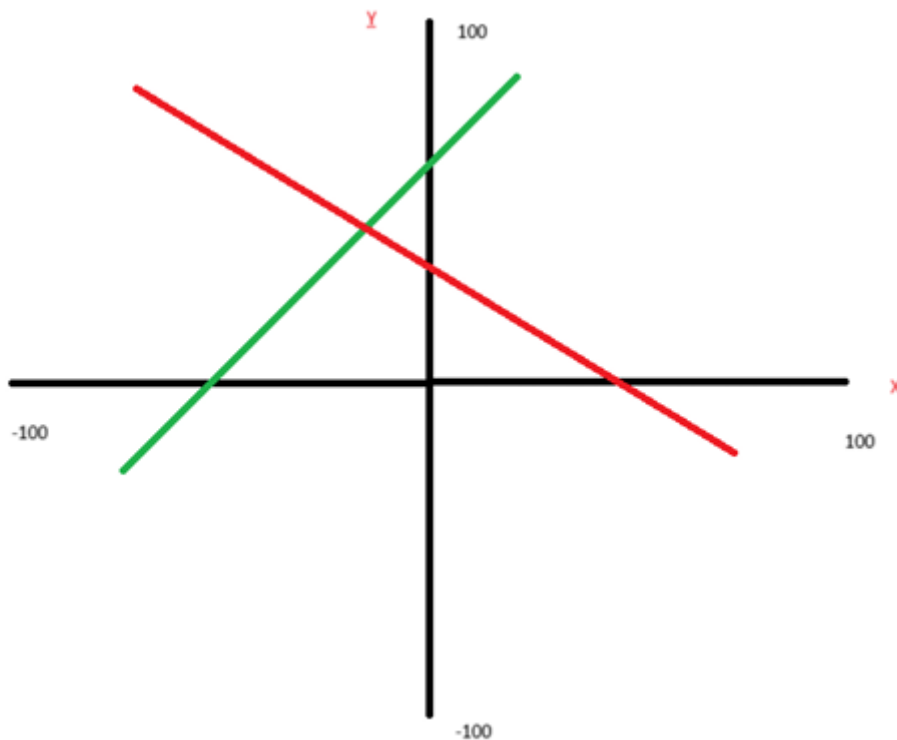
Semester: **FALL 2025-2026**

MID Term LAB Task-2

Student ID: **23-51773-2**

DDA and Integer Pixel Grid

1. Construct a OpenGL Program containing the following conditions:
 - a. Use a separate init() function to declare your own co-ordinate system. Use the range (-100, 100, -100, 100).
 - b. Construct a display function using DDA Algorithm to generate a straight line with input: X1, Y1 (initial vertex) and Xn,Yn (Final Vertex). Consider all three conditions of the DDA Algorithm while implementing the algorithm.
 - c. Create two lines as given below using DDA algorithm:



OpenGL Code

```
#include <windows.h>
#include <iostream>
#include <cmath>
#include <algorithm>
#include <GL/gl.h>
#include <GL/glut.h>
using namespace std;

float a1,b1,a2,b2,c1,d1,c2,d2;

void drawAxes(){
    glColor3f(0,0,0);
    glBegin(GL_LINES);
    glVertex2i(-100,0); glVertex2i(100,0);
    glVertex2i(0,-100); glVertex2i(0,100);
    glEnd();
}

void dda(float X1,float Y1,float X2,float Y2,int r,int g,int b){
    float m=(Y2-Y1)/(X2-X1);
    glColor3ub(r,g,b);
    glBegin(GL_POINTS);
    if(X1>X2){swap(X1,X2);swap(Y1,Y2);m=-m;}
    if(Y1>Y2){swap(Y1,Y2);m=-m;}
    if(m<=1){
        while(X1<X2 && Y1<Y2){
            X1=X1+1;
            Y1=Y1+m;
            glVertex2i(round(X1),round(Y1));
        }
    } else{
        while(X1<X2 && Y1<Y2){
```

```

        X1=X1+(1/m);

        Y1=Y1+1;

        glVertex2i(round(X1),round(Y1));
    }
}

glEnd();
}

void display(){
    glClear(GL_COLOR_BUFFER_BIT);

    drawAxes();

    dda(a1,b1,a2,b2,255,0,0);

    dda(c1,d1,c2,d2,0,255,0);

    glFlush();
}

void init(){
    glClearColor(1,1,1,0);

    glPointSize(4);

    glMatrixMode(GL_PROJECTION);

    glLoadIdentity();

    gluOrtho2D(-100,100,-100,100);
}

int main(int c,char** v){
    cout<<"Line 1 (X1 Y1 X2 Y2): "; cin>>a1>>b1>>a2>>b2;

    cout<<"Line 2 (X1 Y1 X2 Y2): "; cin>>c1>>d1>>c2>>d2;

    glutInit(&c,v);

    glutInitDisplayMode(GLUT_SINGLE|GLUT_RGB);

    glutInitWindowSize(600,600);

    glutCreateWindow("DDA");

    init();

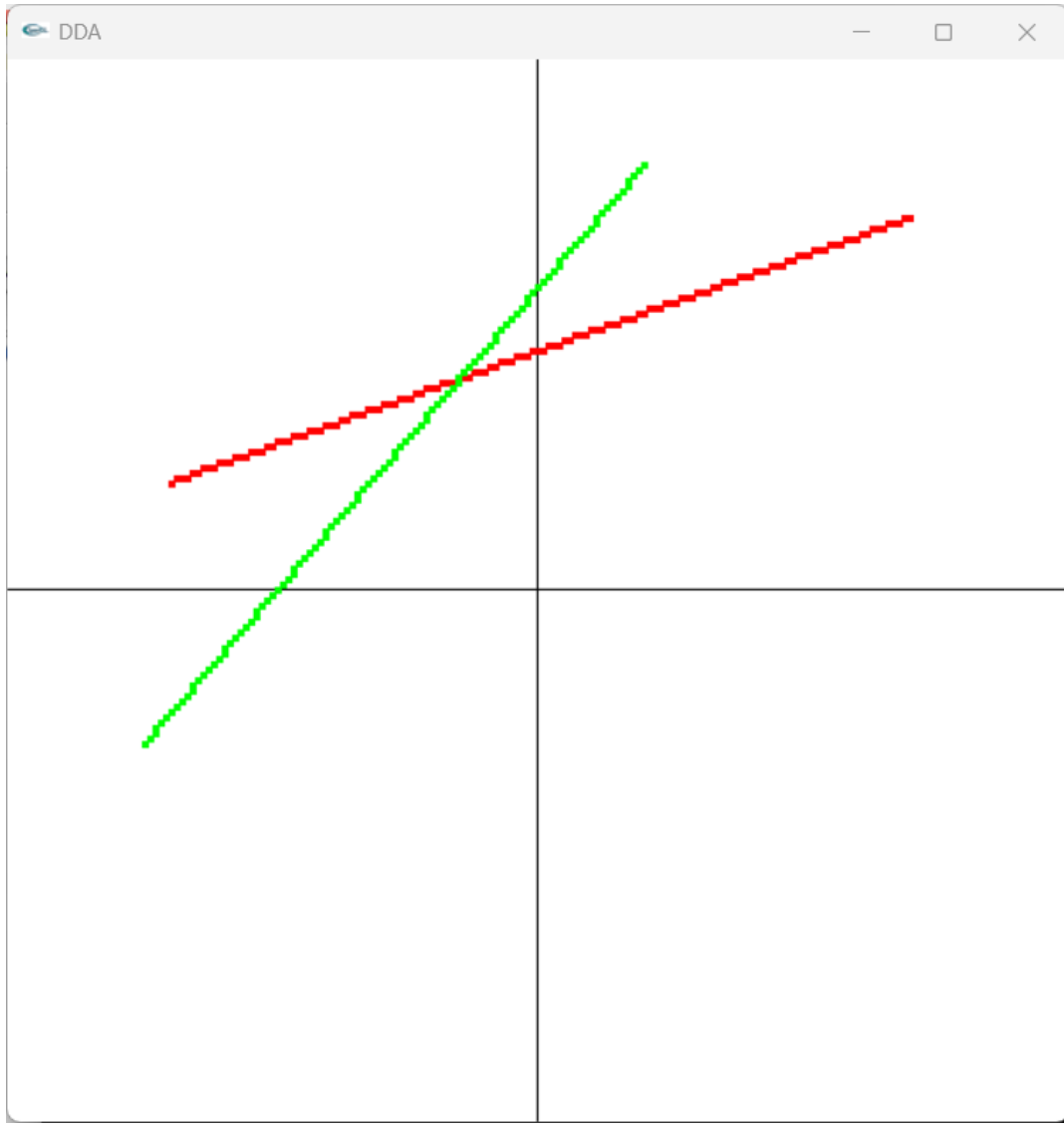
    glutDisplayFunc(display);

    glutMainLoop();

    return 0;
}

```

Output



THE END